

Wandering Spirit: Exploring Cooperative Mixed-Reality Gameplay with Shared Physical Props

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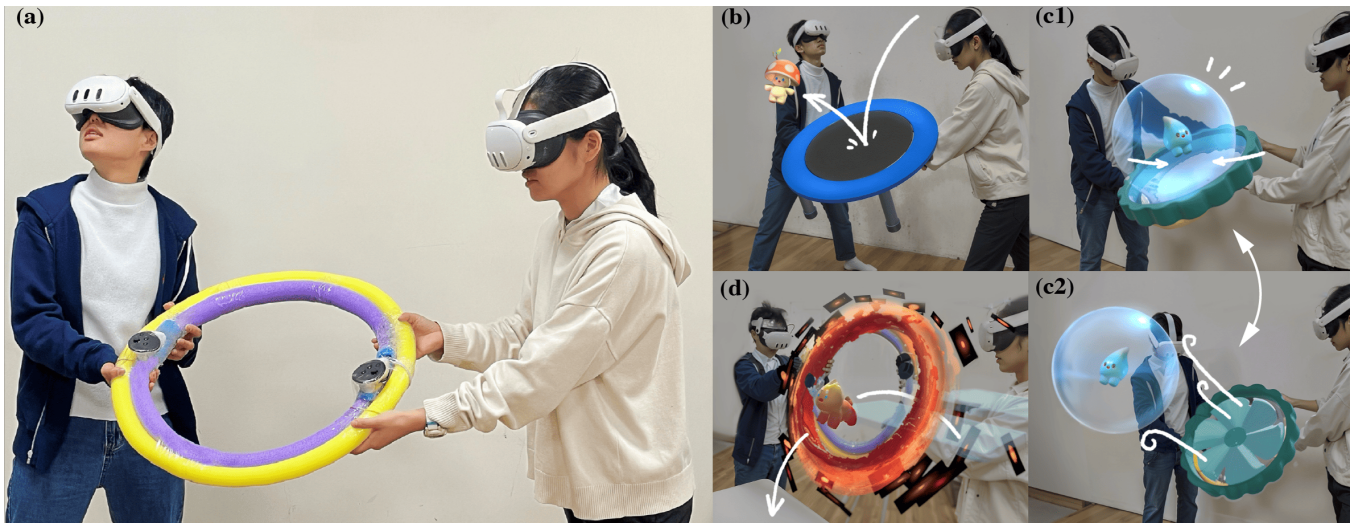


Figure 1: Wandering Spirit: A mixed reality game where two players collaboratively control a physical flexible loop to rescue an elemental spirit through three distinct stages.

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Abstract

Wandering Spirit is a mixed-reality game that fosters collaboration through the shared use of a physical prop. The game's core mechanic centers on a flexible loop manipulated simultaneously by two co-located players, serving as a bridging element between physical and virtual interactions. Through three distinct game stages, the loop transforms into different virtual tools—a trampoline, a bubble/wind generator, and a fiery hoop—each requiring precise synchronization between players to interact with virtual elemental spirits. The shared prop creates natural constraints that guide player coordination while maintaining individual agency. Initial observations suggest that the physical sharing of a single prop creates emergent social dynamics and enhanced spatial awareness between

players, contributing to both mechanical cooperation and social presence. The game demonstrates how shared physical props can serve as effective mediators between physical and virtual spaces, potentially opening new design opportunities for embodied collaborative experiences in mixed reality.

CCS Concepts

• **Human-centered computing** → **Interaction design**; **Mixed reality**.

Keywords

Mixed Reality, Shared Haptic Controller, Co-Located Collaboration, Tangible Interaction

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1 INTRODUCTION

Recent advances in mixed reality (MR) technology, such as Meta Quest 3 and Apple Vision Pro, along with accessible development tools, enable immersive gaming that seamlessly blends virtual and physical environments [1, 22]. Multiplayer games especially benefit, fostering new ways for players to interact and collaborate in shared spaces [8]. Examples include Meta's *Home Sports*, where players engage in physical games like badminton, and *Spatial Ops*, which transforms a room into a dynamic multiplayer battlefield¹.

To enhance immersion in VR and MR settings, previous research has explored alternative controllers and haptic feedbacks [3, 16, 20, 21]. These include handheld controllers that simulate surfaces, textures, and forces [2, 26], devices for grasping and touching [7, 13], and tools providing feedback for weight shifts [6, 28]. However, the use of shared physical props as controllers, particularly for two-player interactions in mixed reality environments, remains largely unexplored. Furthermore, most MR games focus on sports, competitive play, or shooting, with few emphasizing cooperative problem-solving by two players. To address this gap, we propose *Wandering Spirit*, a co-located MR game where two players must coordinate and collaborate using a shared physical controller.

The game provides an engaging experience where players collaborate to guide elemental spirits to their destinations. Its intuitive mechanics ensure accessibility for players of all skill levels. The controller, a donut-shaped loop made from two pool noodles, is easy to acquire and assemble (Figure 1-a). Across three stages, the loop dynamically transforms into various virtual objects, requiring players to coordinate actions like bouncing, squeezing, and rotating to achieve their goals (Figure 1-b, c1, c2, d).

The design objectives of the game are as follows: 1) Explore the potential of shared physical props as two-player controllers. 2) Evaluate their impact on teamwork, engagement, coordination, and immersion. 3) Investigate physical-digital interactions in co-located mixed reality gaming.

¹Spatial Ops, <https://www.resolutiongames.com/spatialops>

2 RELATED WORK

2.1 Haptic Controllers for Collaborative Gaming

Commercial products and research have introduced various haptic controllers to enhance immersion in virtual environments [17, 23, 24]. For example, Nintendo's *Ring-Con* combines dynamic resistance and vibrations to create a tactile and immersive gaming experience². For multiplayer experiences, most consoles, like PlayStation and Nintendo Switch, provide individual haptic feedback through separate controllers, which do not account for co-player interactions [11]. Some studies have explored shared control systems to improve engagement. For instance, *PairPlayVR* demonstrated shared character control, showing improvements in collaborative experiences [30, 31]. Ganesh et al. found that physical collaboration enhances motor performance and coordination [9], while Hashiura et al. showed that shared haptic feedback fosters mutual understanding and collaboration when manipulating a single virtual object [11]. Based on these findings, this project examines how sharing a single control device impacts collaborative gaming, aligning with the broader perspective presented in *Playful Wearables* [5], which highlights the potential of embodied interfaces to foster interaction and unconventional engagement. By exploring shared control dynamics, our work expands the design space of collaborative and embodied interactions beyond traditional paradigms.

2.2 Collaborative Gameplay in Co-Located Mixed Reality

Our game draws inspiration from the popular two-player game *It Takes Two*³, where players collaborate in a shared virtual space by controlling unique characters to navigate challenges. We reimagine this concept for MR, enabling players to act as themselves in the same physical space. Prior works have explored collaborative MR experiences. *Brick[4]* introduced a co-located AR game with a design model for synchronous multiplayer interactions. *Blocks[10]* enabled users to collaboratively create persistent virtual structures, analyzing the impact of time and location on collaboration. *MOFA* [15] proposed an asymmetric MR strategy, assigning distinct roles and perspectives to players, demonstrated through four prototypes. While previous studies focus on design paradigms or static gameplay in MR, our game introduces a dynamic, easy-to-play experience where two players move through the environment, coordinate actions, and collaboratively control a shared physical prop, opening new possibilities for MR gameplay.

3 INTERFACE INNOVATION

Wandering Spirit integrates a shared physical prop into a co-located MR environment. Unlike conventional haptic controllers or individual devices, this prop requires two players to dynamically coordinate movements, fostering teamwork and synchronized actions.

The shared controller is simple and versatile, inspired by prior work on passive haptic feedback using everyday objects [18, 27, 29]. Made from two pool noodles connected into a double-layer loop with VR controllers attached on opposite sides (Figure 2),

²Ring-Con, <https://ringfitadventure.nintendo.com/>

³It Takes Two, <https://www.ea.com/games/it-takes-two>

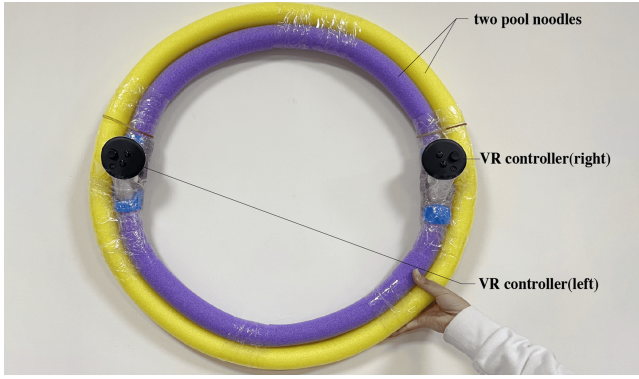


Figure 2: The design of the shared controller, a flexible loop crafted from two pool noodles, with a pair of VR controllers from a single device attached on opposite sides.

it simulates various virtual objects: a trampoline, bubble maker, wind blower, or fiery hoop. Drawing on concepts such as prop-oriented world rotation [25], which leverages real-world props to align passive haptic feedback with virtual objects, the system uses controller position and orientation to track interactions. By aligning players' virtual and physical spaces, the system enables seamless shared control of a single virtual object, enhancing co-located MR collaboration.

In summary, this project combines physical teamwork with virtual interaction, requiring players to control a shared object collaboratively. It pushes the boundaries of MR gaming by encouraging dynamic player movement and interaction, fully leveraging the potential of co-located gameplay to create a more engaging and immersive experience.

4 GAME CONCEPT AND DESIGN

Two players collaborate using the loop to complete three stages—Earth, Water, and Fire (Figure 1)—in a 5m × 5m space. Elemental spirits emerge from multiple starting points, and players must move continuously to prevent them from falling, guiding them to a transfer area and eventually to Utopia. The stage is completed by rescuing a specific number of spirits. Each stage introduces unique mechanics for controlling and manipulating the loop to interact with the spirits. To help players understand how to operate the loop in each stage, a hand-drawn instructional guide is displayed at the beginning of each stage to illustrate the functions of the loop and its effects.

4.1 Stage Design

4.1.1 Earth. To study synchronized physical movement with the shared prop, we designed the Earth stage, where the loop acts as a virtual trampoline inspired by the classic game *Fire* (Figure 1-b, 4-a). Earth spirits spawn at designated wall points (Figure 3-a), signaled by a red light (Figure 3-d), and an audio signal. The players must move continuously, controlling the position and angle of the loop to bounce the earth spirits towards the transfer area (Figure 3-a1). The system tracks the loop's orientation and movement via attached controllers, translating inputs into the virtual environment. Success

depends on synchronized movements and spatial awareness to guide spirits accurately.

4.1.2 Water. The water stage explores more complex cooperation through mode-switching mechanics and coordinated actions. The loop serves as both a bubble maker and a wind blower. The water spirits spawn from a pipe, dropping at its end (Figure 3-b). The players squeeze the loop to create bubbles that encapsulate the water spirits (Figure 1-c1, 4-b), intuitively linking the physical and virtual interactions. Rotating the loop along its diameter axis twice switches it to wind blower mode (Figure 4-c), allowing players to propel bubbles briefly (Figure 1-c2) before it reverts to bubble-making mode. Rotation acts as a mechanism to 'power' the wind blower, adding a layer of strategy and timing. The system detects squeezing by measuring the distance of the controller and identifies rotation through the normal vector of the loop. The players must carefully coordinate mode switching, adjust wind angles, and time actions to guide bubbles to the transfer area before they burst (Figure 3-b1), focusing on precise communication and teamwork.

4.1.3 Fire. The fire stage integrates real-world objects into gameplay, with fire spirits traversing platforms represented by physical objects like tables, creating a tangible link between the physical and virtual realms. The loop becomes a fiery hoop that players use to guide spirits as they jump between platforms (Figure 3-c). Spirits will only leap through the hoop once it is ignited (Figure 1-d). To ignite the hoop, players spin the loop along its normal vector twice, mimicking the sensation of spinning a flaming hoop (Figure 4-d). The system detects rotation by analyzing changes in the relative heights of the controllers. Players must align the hoop precisely to guide spirits in the correct direction (Figure 3-c1), emphasizing seamless integration of physical and virtual elements.

4.2 Implementation Details

The game was developed using Unity with the Meta XR All-in-One SDK, optimized for two Meta Quest 3 devices. The SDK's shared spatial anchors [12, 14] align virtual and physical environments, while the passthrough enables natural player interaction within the shared space. Multiplayer functionality is powered by Photon Fusion 2's shared mode, ensuring reliable server-based interactions. We implemented real-time physical prop tracking by continuously receiving and processing controller location and rotation data to dynamically trigger contextual virtual effects.

During implementation, we faced two technical challenges. First, shadow artifacts on virtual objects caused discomfort and motion sickness. Disabling Screen Space Ambient Occlusion (SSAO) resolved this issue. Second, device calibration made aligning virtual and physical spaces difficult, initially causing synchronization issues. We addressed this by (1) setting the spatial anchor tracking origin to "floor" to reduce height discrepancies and (2) requiring both players to start from the same physical location to prevent alignment errors. For synchronization, covering level transitions, animations, effects, and scoring, we carefully managed network variables and behaviors to ensure consistency.

All character art, including modeling, texturing, and animation, was created in-house. Environmental scenes were primarily blocked out using licensed Unity Asset Store assets. The opening animations

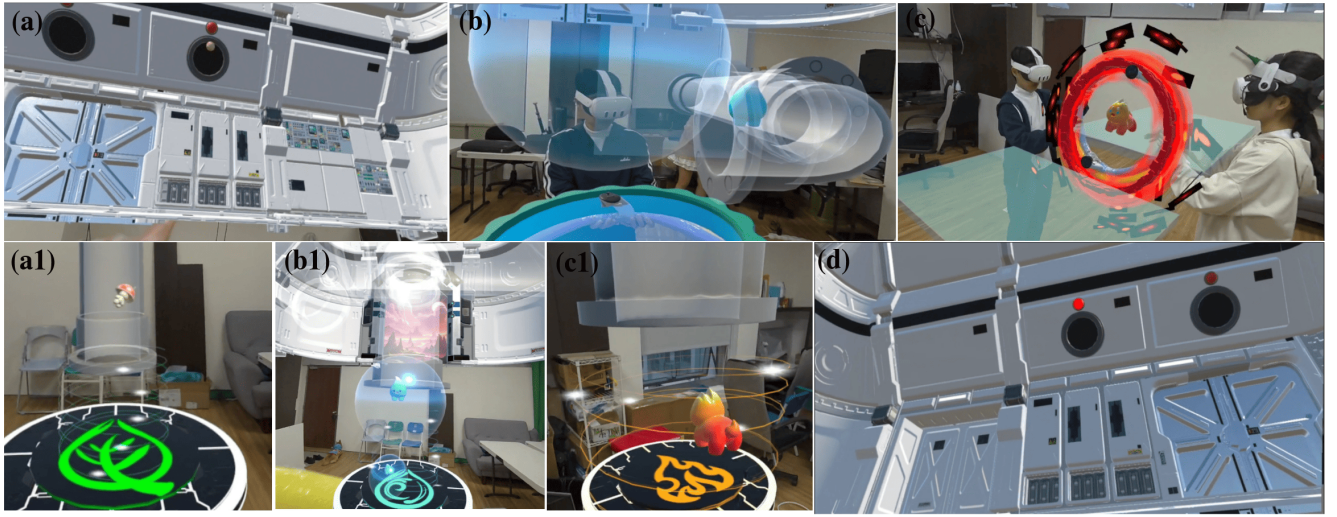


Figure 3: In-game Images of Wandering Spirit: (a) Earth spirit spawn point (a1) Earth stage goal area (b) Water spirit spawn point (b1) Water stage goal area (c) Fire spirit jumping through activated fiery hoop (c1) Fire stage goal area (d) Red light signaling imminent earth spirit spawn

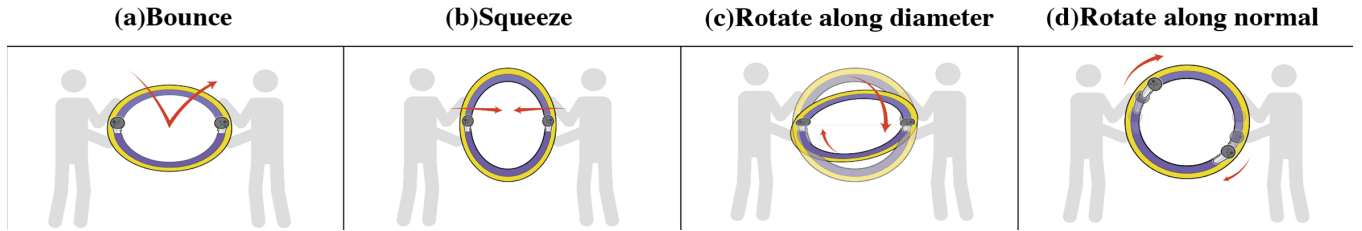


Figure 4: Loop Operations in Wandering Spirit

were hand-drawn, and all sound effects were custom-designed by our team, ensuring a cohesive and original audiovisual experience.

5 PLAYER EXPERIENCE EVALUATION

We conducted a playtest with four teams (P1-P8: 4 female, 4 male, aged 22–27), all experienced with VR and haptic feedback. Sessions took place in a 10m × 5m classroom, beginning with a briefing on gameplay mechanics and objectives. All teams completed the three stages successfully, with each session lasting about 15 minutes. After gameplay, participants completed a 7-point Likert scale questionnaire evaluating immersiveness, focus, enjoyment, ease of control, cooperation, and motion sickness. This was followed by semi-structured interviews for qualitative feedback on cooperation, haptics, MR, and their questionnaire responses. Each session lasted approximately 30 minutes.

The results indicate high levels of enjoyment ($M = 6.13$, $SD = 0.60$), focus ($M = 6.63$, $SD = 0.48$) and satisfaction with cooperative experience ($M = 5.5$, $SD = 0.71$). All participants agreed that cooperative gameplay was very enjoyable and uplifting, a sentiment that was evident during their gameplay sessions. Most attributed their high focus to the need to simultaneously track “the other player’s actions (P6)” and “the object in their hands (P5)”. They agreed that

collaborative control of the loop strengthened their sense of cooperation, such as “Because we had to react quickly to save spirits, we needed to come up with simple and clear commands to communicate with each other (P8)”, and “We have to coordinate our actions to squeeze the bubble, which really requires teamwork and mutual understanding (P4)”. However, participants also highlighted challenges like communication issues, “We kept saying ‘there,’ but it wasn’t clear where ‘there’ actually was (P3)”, and “The other player’s actions didn’t match what I expected (P1)”. Coordination difficulties were also mentioned: “The force on both sides feels uneven, and it seems like the other player has more control (P2)”.

The results also demonstrate high levels of immersiveness ($M = 5.75$, $SD = 1.09$) and ease of control ($M = 5.13$, $SD = 0.60$). Most participants felt the loop effectively simulated virtual props, enhancing interaction with the virtual world. “Especially when squeezing the loop to blow bubbles, the level of pressure really feels just like the actual squeezing motion in your hands (P1)”. Another mentioned, “The trampoline stage is very intuitive, with classic gameplay (P5)”. However, some challenges were reported. A participant stated that “Seeing real-world objects decreases the sense of immersion (P2)”. Interestingly, despite high concentration levels, none of the participants experienced motion sickness ($M = 1.63$, $SD = 0.48$), likely

due to the continuous body movement reducing motion sickness in virtual environment [19].

6 DISCUSSION

Throughout the development of Wandering Spirit, we explored how shared physical props can serve as intuitive controllers in co-located MR gameplay. Our design process showed that physical constraints—such as a shared loop—naturally foster communication and synchronized movement, suggesting that intentionally restrictive shared controls could encourage deeper cooperative engagement.

A key design choice was positioning the virtual environment above the player's eye level, allowing them to stay aware of their real world surroundings. This seamless integration of physical and virtual elements not only maintained immersion but also enabled freer movement within the space. The two-player cooperative design also introduced a built-in safety mechanism: Players could observe their partner's surroundings and help prevent collisions, either through verbal communication or by using the loop to guide them. These insights suggest that shared control in MR gameplay can foster collaborative spatial awareness, informing the design of safer and more accessible MR experiences.

Our design process also raised broader questions about how shared control mechanics shape player dynamics in MR games. Players adopted various collaboration strategies—some took turns leading, while others used verbal or non-verbal cues to synchronize actions. This highlights an opportunity for future research to explore how different shared controller designs influence communication, teamwork, and engagement. Investigating alternative prop materials, adaptive inputs, or asymmetric control schemes could further refine the balance between physical and virtual collaboration, expanding the possibilities of immersive MR gameplay.

However, several technical challenges remain. Player 2 experiences lag, and tracking fails if controllers move outside Player 1's field of view. Calibration inaccuracies also prevent perfect alignment of the loop with virtual objects. Future work could address these issues by replacing VR controllers with computer vision-based pose estimation for the loop, enabling smoother and more seamless interactions.

Another limitation is the lack of depth information on the Meta Quest platform, causing incorrect occlusion where virtual objects always appear in front of the player, limiting virtual environment complexity. To mitigate this, we positioned players face-to-face and moved the virtual scene above their heads to reduce visual obstructions. Incorporating advanced depth detection could resolve occlusion issues, enable proper alignment between virtual and physical objects, and allow for diverse user arrangements, enhancing interaction dynamics and gameplay possibilities.

7 CONCLUSION

In Wandering Spirit, we demonstrated how shared physical props can enhance collaboration, engagement, and immersion in co-located mixed-reality environments. The playtest revealed high levels of enjoyment, focus and cooperative satisfaction, with shared control that fosters natural communication, coordination, and teamwork. These findings underscore the potential of shared physical props as

intuitive controllers that balance collaborative challenges with engaging gameplay, providing valuable insights for MR game design. Future work should explore a wider range of shared prop designs, refine tracking and occlusion issues, and examine how these interaction principles can extend beyond gaming, enabling MR experiences like Wandering Spirit to further push the boundaries of immersive and collaborative interaction design.

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